1.6.2 Calculation of Molecular Formula from Empirical Formula:

Compound	Empirical Formula	Molar mass	Empirical Formula mass	Whole number (n)	Molecular formula
Acetic acid	CH_2O	60	30	$\frac{60}{30} = 2$	$(CH_2O) \times 2$ $C_2H_4O_2$
Hydrogen Peroxide	HO	34	17	$\frac{34}{17} = 2$	$(HO) \times 2$ H_2O_2
Lactic acid	CH_2O	90	30	$\frac{90}{30} = 3$	$(CH_2O) \times 3$ $C_3H_6O_3$
Tartaric acid	$C_2H_3O_3$	150	75	$\frac{150}{75} = 2$	$(C_2H_3O_3) \times 2$ $C_4H_6O_6$
Benzene	CH	78	13	$\frac{78}{13} = 6$	$(CH) \times 6$ C_6H_6

Let us understand the calculations of molecular formula from the following example.

Two organic compounds, one present in vinegar (molar mass: 60 g mol^{-1}), another one present in sour milk (molar mass : 90 g mol^{-1}) have the following mass percentage composition. C-40%, H-6.6% ; O-53.4%. Find their molecular formula.

Since both compounds have same mass percentage composition, their empirical formula are the same as worked out in the example **problem no 2** . Empirical formula is CH_2O. Calculated empirical formula mass (CH_2O) = 12 + (2×1) + 16 = 30 g mol^{-1}.

Formula for the compound present in vinegar

$$n = \frac{\text{Molar mass}}{\text{calculated empirical formula mass}} = \frac{60}{30} = 2$$

$\therefore$ Molecular formula $= (CH_2O)_2$

$= C_2H_4O_2$

(acitic acid)

Calculation of molecular formula for the compound present in sour milk.

$$n = \frac{\text{Molar mass}}{30} = \frac{90}{30} = 3$$

Molecular formula $= (CH_2O)_3$

$= C_3H_6O_3$

(lactic acid)

Evaluate Yourself

6) Experimental analysis of a compound containing the elements x,y,z on analysis gave the following data. x = 32 %, y = 24 %, z = 44 %. The relative number of atoms of x, y and z are 2, 1 and 0.5, respectively. (Molecular mass of the compound is 400 g) Find out.

i) The atomic masses of the element x,y,z.

ii) Empirical formula of the compound and

iii) Molecular formula of the compound.

1.7 Stoichiometry

Have you ever noticed the preparation of *kesari* at your home? In one of the popular methods for the preparation of *kesari*, the required ingredients to prepare six cups of *kesari* are as follows.

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